

The Christoffel Rank Read

Analytic Rank and Low Zeros from Frobenius Traces Alone

Arithmetic-Moment Stieltjes Pipeline — Measured Results, v2

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Abstract

This note records what the arithmetic-moment instrument reads that is true, measured, and externally anchored. **The central results:** (i) the Gaussian-quadrature weight at the spectral origin equals the analytic rank as an exact integer — 0, 1, 2, 3, 4 with no residual on exact measures; (ii) with the explicit-formula conventions pinned by a hard gate on the Riemann zeta function (agreement with direct zero sums to 10^{-37}), the same read executes on real elliptic curves from point counts alone, returning the full rank ladder 0, 1, 2, 3 across conductors 11 to 5077, with $w_0 = 2.00000002$ on 389a1; (iii) the low zeros come out with the ranks — 389a1's first zero to every published digit, the first Riemann zero to eight digits from primes; (iv) the read's resolution is governed by a simple density law, $\sqrt{(Kt)}$, verified by prediction. One earlier channel is killed and kept for what its failure located: the cone-approach decay rate carries no rank. Everything below is at the status the code earned.

1. The Instrument

The pipeline takes local arithmetic data and produces the low-lying spectrum of an L -function. Its skeleton is classical moment theory, and every stage is a named object:

$\{a_p\} \rightarrow \{\mu_k\} \rightarrow H(\text{Hankel}) \rightarrow L(\text{Cholesky}) \rightarrow J(\text{Jacobi}) \rightarrow (\lambda_i, w_i)$

Local traces of Frobenius a_p feed the even moments μ_k of a symmetric, Gaussian-windowed measure on the zeros, via Weil's explicit formula. The moments populate a Hankel matrix H ; its Cholesky factor yields the three-term recurrence of the measure's orthogonal polynomials; the associated Jacobi matrix J is tridiagonal with zero diagonal — forced by the y^2 symmetry, the functional equation in operational form; and the eigenpairs of J are the Gauss quadrature nodes and weights. The nodes are the imaginary parts of the zeros; the weights are the spectral masses. Everything below concerns one node — the origin — and its weight w_0 , plus the fidelity of the rest of the spectrum.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality

2. Confirmed: The Origin Weight Is the Rank (Exact Measures)

An L -function of analytic rank r has a zero of order r at the center — a point mass of weight r at the origin of the spectral measure. On measures with a known central atom and exact moments, the quadrature recovers this mass as an exact integer:

Constructed rank r	Recovered origin weight w_0	Quadrature size
0	4.12×10^{-58} (machine zero)	11
1	1.0000000	11
2	2.0000000	11
3	3.0000000	11
4	4.0000000	11

Exact measures, Gaussian aperture $t = 0.15$, dps 60. Rank 0 returns 10^{-58} — the read reports no mass at the origin, correctly.

The rank is never approached by evaluating L or its derivatives at a point — the classical search operation. The moments settle, positivity routes them into a quadrature rule, and the integer is the share of the conserved total mass parked at the fixed point. The conservation is what makes the read trustworthy: at every truncation depth $\sum w_i = \mu_0$ to machine precision, so an integer atom cannot leak into the continuum. *Balance reads as absence for rank 0; an integer atom reads as that integer for rank ≥ 1 .*

3. The Calibration Closed: The ζ Gate

Running the read on real curves requires the explicit formula's conventions to be exact — every constant, every transform factor. The derivation located two defects in the prior arithmetic realization: the prime side was missing the factor $(-1)^k(4t)^{-k}$ from the Fourier transform of the moment test function (the Hermite prefactor, growing with moment index — which is why the distortion was k -dependent, not a single scalar), and the archimedean bracket was at half strength.

The corrected formula was then put through a hard gate: compute the moments of the Riemann zero measure two independent ways — from primes via the explicit formula, and directly as sums over the actual zeros — and demand agreement. The gate requirement was 10^{-10} . The result:

Moment k	From zeros (direct sum)	From primes (explicit formula)	Relative error
0	0.0000917567009714714854	0.0000917567009714714854	4.4×10^{-37}
2	3.66265994456063275	3.66265994456063275	4.4×10^{-40}

Moment k	From zeros (direct sum)	From primes (explicit formula)	Relative error
4	146214.944127959822	146214.944127959822	1.7×10^{-40}
6	5839354339.43087113	5839354339.43087113	3.4×10^{-41}
8	233672528343983.526	233672528343983.526	6.9×10^{-41}

$t = 0.05$, dps 40, primes to 800. Worst error across $k = 0 \dots 8$: 4.4×10^{-37} . Gate requirement was 10^{-10} ; it passed by twenty-seven orders of magnitude.

One structural fact inside this table deserves its own sentence. The pole term of ζ contributes ≈ 2.025 to the zeroth moment; the true moment is 0.0000918. The archimedean and prime sides cancel the pole to forty digits — a fully determined system whose enormous internal tensions net out to a whisper. **The wash, measured.**

With conventions pinned, the Jacobi read was pointed at the prime-side moments of ζ itself:

Riemann zero (true)	Recovered from primes	Digits
14.1347251417...	14.13472517	8
21.0220396...	21.0220673	5
25.0108575...	25.0120974	4
30.4248761...	30.5754444	edge

$t = 0.02$, $K = 10$, primes to 1200. No zero used as input anywhere. Resolution degrades gracefully toward the aperture edge, as the frontier behavior requires.

4. Rank from Point Counts on Real Curves

The identical machinery ports to elliptic curves with three substitutions: the archimedean bracket becomes $\log N - 2 \log 2\pi + 2 \operatorname{Re} \psi(1 + ir)$, the pole term is deleted (the L-function is entire), and the von Mangoldt weights become the unit-normalized Frobenius recurrence $u_0 = 2$, $u_1 = a_p/\sqrt{p}$, $u_m = u_1 u_{m-1} - u_{m-2}$, with bad primes handled by nonsingular point counts. The ψ -integral is curve-independent, computed once, shared by every curve.

The four canonical curves, ranks read from Frobenius traces alone — no L-value, no zero, no rank supplied anywhere in the pipeline:

Curve	Conductor	True rank	Read w_0	Settings
11a1	11	0	7.95×10^{-12}	$t=0.10$, $K=12$

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Curve	Conductor	True rank	Read w_0	Settings
37a1	37	1	1.0000001	t=0.10, K=12
389a1	389	2	2.00000002	t=0.15, K=20
5077a1	5077	3	3.000099	t=0.15, K=20

Rank ladder 0–3 complete. Rank 0 reads machine zero; rank 2 reads to eight digits.

And the low zeros come out with the ranks, anchored against the published digits carried in the session record:

Curve	Recovered nodes	External anchors (LMFDB digits)
37a1	5.00326, 6.8779, 8.0557	5.00317, 6.87039, 8.01433
389a1	2.8761, 4.41695, 5.79526	2.8761, 4.4169, 5.7934
5077a1	2.05355, 3.27301	2.05247, 3.26244
11a1	6.36261, 8.60356, ...	(internal prediction)

389a1's first zero matches every published digit. These are the fingerprints that discriminate sibling newforms of the same conductor.

5. The Density Law: $\sqrt{(Kt)}$ Governs the Read

The resolution of the origin read is set by the quadrature's node density inside the visible window. The window reaches to $\gamma \approx \sqrt{(K/t)}$; the rule supplies $K+1$ nodes; the density is therefore $\propto \sqrt{(Kt)}$. This was extracted from one failed refinement (Section 7.4) and then tested by prediction: raise the density and the rank read must tighten. It did:

Run	$\sqrt{(Kt)}$	w_0 (389a1, true 2)	w_0 (5077a1, true 3)
t=0.06, K=16	0.98	2.0123	3.1497
t=0.10, K=12	1.10	2.0025	3.0699
t=0.15, K=20 (predicted best)	1.73	2.00000002	3.000099

The prediction was stated before the run: density 1.73 should beat both prior settings. It improved the rank-2 read by five orders of magnitude.

This is the aperture-frontier behavior rendered as a design rule: budget spent without raising $\sqrt{(Kt)}$ is budget wasted; budget spent raising it converts directly into digits of the invariant.

6. Killed: The Decay Rate Does Not Encode Rank

The Hankel's minimum eigenvalue decays geometrically toward the positivity wall. An earlier reading held that this decay rate was a second, dynamical rank channel (a correlation near -0.97 on three curves), later softened to "the rate's rank-ordering is the invariant." **Tested on exact measures where rank is the only variable, both forms are false:** the slope and its ordering scramble arbitrarily with aperture — $(1,3,2,0)$, $(0,2,3,1)$, $(3,2,1,0)$, $(0,1,2,3)$ across four apertures. The -0.97 was three curves, two apertures, and 3–4 fit points per slope: an artifact of small numbers.

Killing it locates rank precisely: rank lives in the **static weight at the origin node** and nowhere in the dynamics of the wall approach. The instrument has exactly one rank channel, and Sections 2 and 4 are it.

7. Productive Errors

Places where the wrong thing was done and the result was still informative — did A, got B, and B was the finding. Ordinary bugs are omitted; only errors that revealed structure are kept.

7.1 The wall that would not recede

A shallow wall was blamed on precision, so precision was raised — dps 30 to 160. **The wall did not move.** The failed fix was the finding: the wall is a genuine positivity event in the moment sequence, a mathematical boundary, not a numerical floor. Had precision cured it, a mathematical boundary would have been mislabeled as an instrument limit.

7.2 The clock with no atom at its center

An early archimedean term — conductor factor only, digamma omitted — produced healthy-looking moments whose spectrum contained **no node at the origin** for the rank-3 curve 5077a1. The instrument refused to hallucinate the atom, and the refusal drew the line: the origin mass is carried by the archimedean–prime balance, not by prime data alone. Rank is not a decoration on a generic measure.

7.3 Synthetic exactness versus arithmetic calibration

With the digamma restored but the transform factor still missing, real-curve reads came back monotone in rank but scaled — 0.001, 0.26, 0.64 for ranks 0, 1, 2 — while the identical read on exact synthetic measures returned 0, 1, 2, 3, 4 with zero error. The contrast separated two statements that a single clean run would have fused: **the method was proven, and the arithmetic realization was miscalibrated** — a labeled gap of known shape. The gap was then closed by derivation and sealed by the ζ gate at 10^{-37} (Section 3).

7.4 The refinement that went backward

To tighten the high-conductor rank reads, budget was added in the natural-seeming direction: wider aperture, more moments ($t = 0.06$, $K = 16$). **The reads got worse** — $2.0025 \rightarrow 2.0123$, $3.070 \rightarrow 3.150$. Doing A and getting the opposite of B exposed the actual resolution variable: not window width, not moment count, but their product — node density $\sqrt{(Kt)}$, which the "refinement" had lowered from 1.10 to 0.98. The corrected move (density 1.73) was stated as a prediction and delivered five orders of magnitude on the rank-2 read (Section 5). The backward step is what turned a tuning knob into a law.

8. Status Ledger

Statement	Status
Origin quadrature weight = analytic rank (exact measures)	CONFIRMED — $0,1,2,3,4$ exact
ζ moments from primes = direct zero sums	CONFIRMED — 4×10^{-37} worst error
Riemann zeros readable from primes alone	CONFIRMED — γ_1 to 8 digits
Origin weight = rank on real curves, from point counts	CONFIRMED — ladder $0,1,2,3$; $389a1$ to 8 digits
Low elliptic zeros match external anchors	CONFIRMED — $389a1 \gamma_1$ to all published digits
Read resolution governed by node density $\sqrt{(Kt)}$	MEASURED — 3-point prediction test
Total mass $\Sigma w_i = \mu_0$ conserved at every depth	CONFIRMED — machine precision
Wall is a positivity / representability event	CONFIRMED — invariant under precision
Decay rate is a rank channel (either form)	KILLED — ordering random by aperture

9. One Line of Framing

A domain that contains all possibilities is the result of the abstract constraints of every impossibility.

The instrument is now a working witness for that sentence at two scales. The positive cone — the space of all realizable measures — is carved out by every positivity constraint that cannot be violated, and the wall is the exact surface where the last admissible configuration gives way. Inside the cone, the one integer the instrument reads perfectly is not written anywhere in the data: it is the mass the constraints force to sit at the single point the functional equation permits no gradient. And the ζ gate showed the same shape in the large — a pole of size 2 annihilated to forty digits by the balance of two infinite sides, structure cancelling below the reader's floor until only the true residue remains. The possibilities the read returns are the settled residue of the impossibilities the cone excludes. The map extends by driving such anchors: ζ , then the rank ladder, then the density law — each one a place where the constraints closed and left an integer or an exchange rate behind.

— end —